



Geopolitical risks and exchange rates

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ABSTRACT

This paper investigates the effects of geopolitical risks on exchange rates of 35 countries. The investigation is based on a structural vector autoregression model, where developments in oil prices, economic activity, inflation and policy rates are controlled for. The empirical results suggest that shocks to geopolitical risks result in currency depreciations only in China, Israel, the Philippines and the United States, whereas they result in currency appreciations mostly in South Africa, Brazil, Australia, and Iceland, among others, after one year. When the heterogeneity across countries is further investigated, it is shown that currencies of countries that are involved more in global value chains depreciate more following shocks to geopolitical risks, especially in the short run. Robustness checks indicate that the latter result is mostly driven by geopolitical acts. Important policy suggestions follow for countries that are integrated into global value chains.

1. Introduction

Geopolitical events can significantly affect exchange rates by influencing international trade flows, investor behavior, and economic prospects (Hossain et al., 2024). This may include trade disruptions due to sanctions or military actions that can disrupt international imports and exports, affecting the demand and supply of a currency and consequently its value (Kisswani and Elian, 2021). This may also include political instability or conflicts leading to capital flows as investors seek safer assets, leading to currency appreciation or depreciation of the affected countries (Ftiti et al., 2024; Yilmazkuday, 2024). Geopolitical events can also alter a country's economic outlook, influencing foreign direct investment and potentially causing currency appreciation or depreciation (Aiyar et al., 2024). Although central banks can play a role by intervening in currency markets to manage volatility, shocks to commodity prices due to geopolitical tensions can have significant effects on exchange rates, particularly in countries that import or export such commodities (Mitsas et al., 2022; Asadollah et al., 2024). Overall, currencies of countries can depreciate or appreciate following geopolitical shocks, where an empirical investigation is required to measure the corresponding effects.

Based on this background, this paper investigates the effects of geopolitical risks on exchange rate of 35 countries for the monthly period between January 2000 and June 2024. The investigation is based on a structural vector autoregression model, where developments in oil prices, economic activity, inflation and policy rates are controlled for. The empirical results suggest that shocks to geopolitical risks result in currency depreciations only in China, Israel, the Philippines and the United States, whereas they result in currency appreciations mostly in South Africa, Brazil, Australia, and Iceland, among others, after one year. When the heterogeneity across countries is further investigated, it is shown that currencies of countries that are involved more in global value chains depreciate more following shocks to geopolitical risks, especially in the short run. The latter result is shown to be driven more by geopolitical acts

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when geopolitical risks are decomposed into acts versus threats.

This paper contributes to the literature along the following lines. First, this paper investigates a sample of 35 countries that makes it possible to analyze the heterogeneity across countries regarding the effects of geopolitical risks on exchange rates. In comparison, studies such as by [Bossman et al. \(2023\)](#) and [Salisu et al. \(2022\)](#) investigate only a limited number of countries. Second, to measure the corresponding effects on exchange rates, this paper investigates the monthly period between January 2000 and June 2024 that cover many different geopolitical events such as the September 11 attacks, the Iraq War, the Russo-Ukraine War, and the Israel-Hamas War. In comparison, studies such as by [Bossman et al. \(2023\)](#) and [Hossain et al. \(2024\)](#) focus on the effects of single geopolitical events. Third, to our knowledge, this is the first study showing that the currencies of countries depreciate more following geopolitical shocks, when they participate more in global value chains. Fourth, this paper shows that the results are mostly driven by geopolitical acts rather than geopolitical shocks, having several policy implications for countries as discussed at the end of this paper.

The rest of the paper is organized as follows. The next section introduces the empirical methodology and the data set. [Section 3](#) depicts the empirical results, where robustness checks are also conducted. [Section 4](#) concludes by providing policy suggestions as well as guidelines for future research.

2. Empirical methodology and data

For the empirical investigation, the structural vector autoregression (SVAR) model is defined as $z_t = (gpr_t, gea_t, oil_t, pol_t, inf_t, nxr_t)'$, where gpr_t is the newspaper-based global geopolitical risk measure borrowed from [Caldara and Iacoviello \(2022\)](#) and normalized to one standard deviation as in [Caldara et al. \(2022\)](#), gea_t is the percentage deviations of global real economic activity from its trend that is borrowed from [Kilian \(2009\)](#), and oil_t represents the growth in global oil prices that are obtained from the Federal Reserve Economic Data as the average of Brent, West Texas Intermediate, and Dubai crude oil prices. The domestic variables are obtained from the Bank of International Settlements, where pol_t represents the changes in domestic policy (interest) rates, inf_t is the domestic inflation defined as the growth of consumer prices, and nxr_t is the percentage change in the nominal effective exchange rates (measured as currency depreciation). Due to data availability, the combined sample covers the monthly period between January 2000 and June 2024 for 35 countries, where the estimations are achieved for each country separately to investigate the heterogeneity across countries.

The formal investigation of the SVAR model is based on the following expression:

$$A_0 z_t = a + \sum_{k=1}^{12} A_k z_{t-k} + u_t \quad (1)$$

where u_t is the vector of serially and mutually uncorrelated structural innovations. For estimation purposes, the model is expressed in reduced form as

$z_t = b + \sum_{k=1}^{12} B_k z_{t-k} + e_t$, where $b = A_0^{-1}a$, $B_k = A_0^{-1}A_k$ for all k , and the number of lags is set to 12 based on the deviance information criterion.¹ It is postulated that the structural impact multiplier matrix A_0^{-1} has a recursive structure such that the reduced form errors e_t can be decomposed according to $e_t = A_0^{-1}u_t$, where shock sizes are standardized to one standard deviation. The recursive structure imposed on A_0^{-1} requires an ordering of the variables for which we utilize the ordering in $z_t = (gpr_t, gea_t, oil_t, pol_t, inf_t, nxr_t)'$. In particular, the global geopolitical risk gpr_t is ordered first as it is assumed to be independent of other variables contemporaneously. The global real economic activity gea_t is ordered before oil prices as in [Kilian \(2009\)](#), as the former is assumed to affect the latter contemporaneously. Domestic policy (interest) rates pol_t are ordered before domestic inflation inf_t due to the monetary policy affecting inflation. Finally, the nominal effective exchange rates nxr_t are ordered the last as the variable of interest in this paper. It is important to emphasize that the results are highly similar for alternative orderings of these variables.

All estimations are achieved by using the BEAR toolbox introduced by [Dieppe et al. \(2016\)](#), where they are achieved by a Bayesian approach with Minnesota priors. This corresponds to generating posterior draws for the structural model parameters by transforming each reduced-form posterior draw. In particular, for each draw of the covariance matrix from its posterior distribution, the corresponding posterior draw for A_0^{-1} is constructed by using by Cholesky decomposition, where shock sizes are standardized to one standard deviation.

In the Bayesian framework, a total of 2000 samples are drawn, where a burn-in sample of 1000 draws is discarded. The remaining 1000 draws are used to determine the structural impulse responses and forecast error variance decomposition measures. While the median of each distribution is considered as the Bayesian estimator, the 16th and 84th quantiles of distributions are used to construct the 68 % credible sets (that is the standard measure considered in the Bayesian literature).

3. Empirical results

This section depicts the empirical results of the SVAR model, where responses of exchange rates to geopolitical risks as well as the contribution of geopolitical risks are depicted. Robustness checks are also achieved in this section to consider alternative measures of geopolitical events as well as the effects of the Russian ruble to investigate the potential impact of the Russo-Ukraine War on other

¹ The deviance information criterion in this framework is superior to others as it is naturally suited for Bayesian analysis, where we work with posterior distributions of parameters. Nevertheless, the results are very similar for alternative numbers of lags.

exchange rates.

3.1. Effects of geopolitical risks on exchange rates

The cumulative impulse responses of exchange rates (represented as percentage depreciation) to one standard deviation of shocks in geopolitical risks are depicted in Fig. 1 and summarized in Table 1 for alternative horizons. Specifically, following one standard

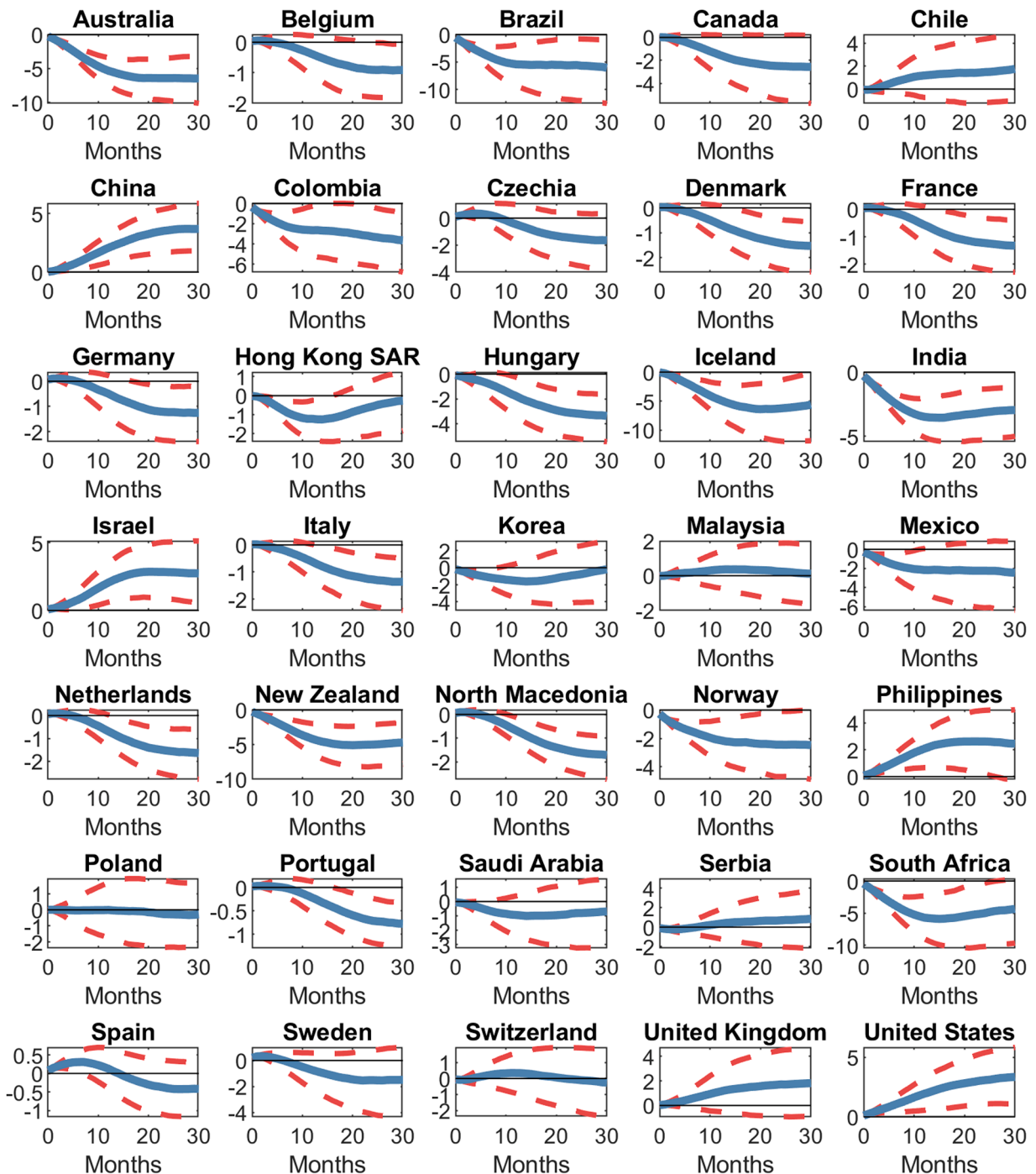


Fig. 1. Responses of exchange rates to geopolitical risks. Notes: Figures represent the cumulative impulse responses of exchange rates (represented as percentage depreciation) to one standard deviation shocks on global geopolitical risks. Solid lines represent the median across 1000 draws, whereas dotted lines represent the 68 % credible sets.

deviation of a shock in geopolitical risks, the nominal effective exchange rates depreciate in China (2.1 %), Israel (2.1 %), the Philippines (2.1 %) and the United States (1.9 %), whereas they appreciate in South Africa (5.7 %), Brazil (5.4 %), Australia (5.3 %), Iceland (4.8 %), New Zealand (4.2 %), India (3.5 %), Colombia (2.7 %), Norway (2.1 %), Hungary (1.8 %), Hong Kong SAR (1.2 %), and North Macedonia (0.7 %) in a statistically significant way one year after a shock to geopolitical risks. It is implied that the effects of geopolitical shocks on exchange rates are highly heterogenous across countries.

The percentage contributions of geopolitical risks to exchange rate volatilities based on the forecast variance decomposition analysis are given in Fig. 2 and summarized in Table 2 for alternative horizons. The results suggest that the exchange rate volatilities of Czechia (5.3 %), Poland (5.2 %), Switzerland (3.4 %), North Macedonia (2.8 %), and United Kingdom (2.8 %) are affected the most, whereas those of Sweden (0.5 %), Mexico (0.5 %), Brazil (0.6 %), Canada (0.7 %) and Serbia (0.8 %) are affected the least by shocks to geopolitical risks. Therefore, there is also heterogeneity across countries regarding the contribution of geopolitical risks to the volatilities in exchange rates.

3.2. Understanding the heterogeneity across countries

In order to understand the reasons behind the heterogeneity across countries, we investigate how the effects of geopolitical risks on exchange rates change with the global value chain participation of countries borrowed from Borin and Mancini (2019) as shown in Tables 1 and 2 in percentage terms. This is motivated by studies such as Bednarski et al. (2024), since participation in global value chains can amplify the effects of geopolitical risks on exchange rates by increasing a country's reliance on international trade and investment flows, making its currency more susceptible to sudden shifts in risk sentiment and capital flight.

The relationship between the global value chain participation of countries and the cumulative response of their exchange rates to geopolitical risks is given in Fig. 3 for alternative horizons after the shock. As is evident, the exchange rates of countries with higher global value chain participation depreciate more by geopolitical shocks, especially in the short run, since the correlation coefficient

Table 1
Exchange rate depreciation after shocks to geopolitical risks.

Country Information			Cumulative Impulse Responses of Exchange Rates			
Country Name	Country Code	GVC Participation	Contemporaneous	After One Month	After Three Months	After One Year
Australia	AU	49.92 %	-0.25*	-0.60*	-1.43*	-5.33*
Belgium	BE	72.28 %	0.04*	0.06*	0.03	-0.39
Brazil	BR	33.98 %	-0.57*	-1.18*	-2.29*	-5.42*
Canada	CA	42.79 %	0.02*	0.01	-0.13	-1.55
Chile	CL	37.53 %	-0.06*	-0.05	0.13	1.18
China	CN	37.34 %	0.02*	0.09*	0.35*	2.08*
Colombia	CO	28.86 %	-0.37*	-0.75*	-1.36*	-2.65*
Czechia	CZ	57.20 %	0.11*	0.19*	0.31*	-0.46
Denmark	DK	59.83 %	0.03*	0.03*	-0.03	-0.67
France	FR	53.53 %	0.04*	0.05*	0.01	-0.57
Germany	DE	56.73 %	0.07*	0.11*	0.10	-0.49
Hong Kong SAR	HK	71.52 %	-0.01*	-0.05*	-0.25*	-1.21*
Hungary	HU	51.38 %	-0.08*	-0.19*	-0.37*	-1.78*
Iceland	IS	48.97 %	-0.04*	-0.21*	-0.91*	-4.82*
India	IN	50.19 %	-0.30*	-0.66*	-1.40*	-3.49*
Israel	IL	47.58 %	0.08*	0.15*	0.34*	2.06*
Italy	IT	47.40 %	0.02*	0.02	-0.04	-0.62
Korea	KR	51.46 %	-0.16*	-0.35*	-0.67*	-1.52
Malaysia	MY	58.58 %	-0.02*	0.00	0.09	0.35
Mexico	MX	39.28 %	-0.27*	-0.52*	-0.98*	-2.16
Netherlands	NL	68.17 %	0.06*	0.09*	0.07	-0.69
New Zealand	NZ	33.62 %	-0.28*	-0.58*	-1.21*	-4.20*
North Macedonia	MK	53.87 %	0.06*	0.10*	0.08	-0.67*
Norway	NO	57.73 %	-0.27*	-0.57*	-1.03*	-2.13*
Philippines	pH	53.03 %	0.07*	0.19*	0.51*	2.08*
Poland	PL	59.41 %	0.01*	0.01	-0.01	-0.01
Portugal	PT	49.99 %	0.02*	0.03*	0.03	-0.22
Saudi Arabia	SA	41.86 %	-0.05*	-0.09*	-0.23*	-0.94
Serbia	RS	60.44 %	-0.10*	-0.18*	-0.23	0.35
South Africa	ZA	42.88 %	-0.42*	-0.97*	-2.08*	-5.70*
Spain	ES	52.14 %	0.08*	0.15*	0.24*	0.13
Sweden	SE	59.98 %	0.21*	0.32*	0.30*	-0.74
Switzerland	CH	47.06 %	-0.06*	-0.08*	-0.03	0.34
United Kingdom	GB	54.43 %	0.03*	0.08*	0.23	1.13
United States	US	39.73 %	0.11*	0.25*	0.55*	1.94*

Notes: GVC Participation stands for the percentage participation in the global value chains. The estimates represent the cumulative impulse responses of exchange rates (represented as percentage depreciation) to one standard deviation shocks on global geopolitical risks. * represents statistically significant estimates based on the 68 % credible sets. The estimations are achieved for each country separately using the structural vector autoregression model for the sample period between January 2000 and June 2024.

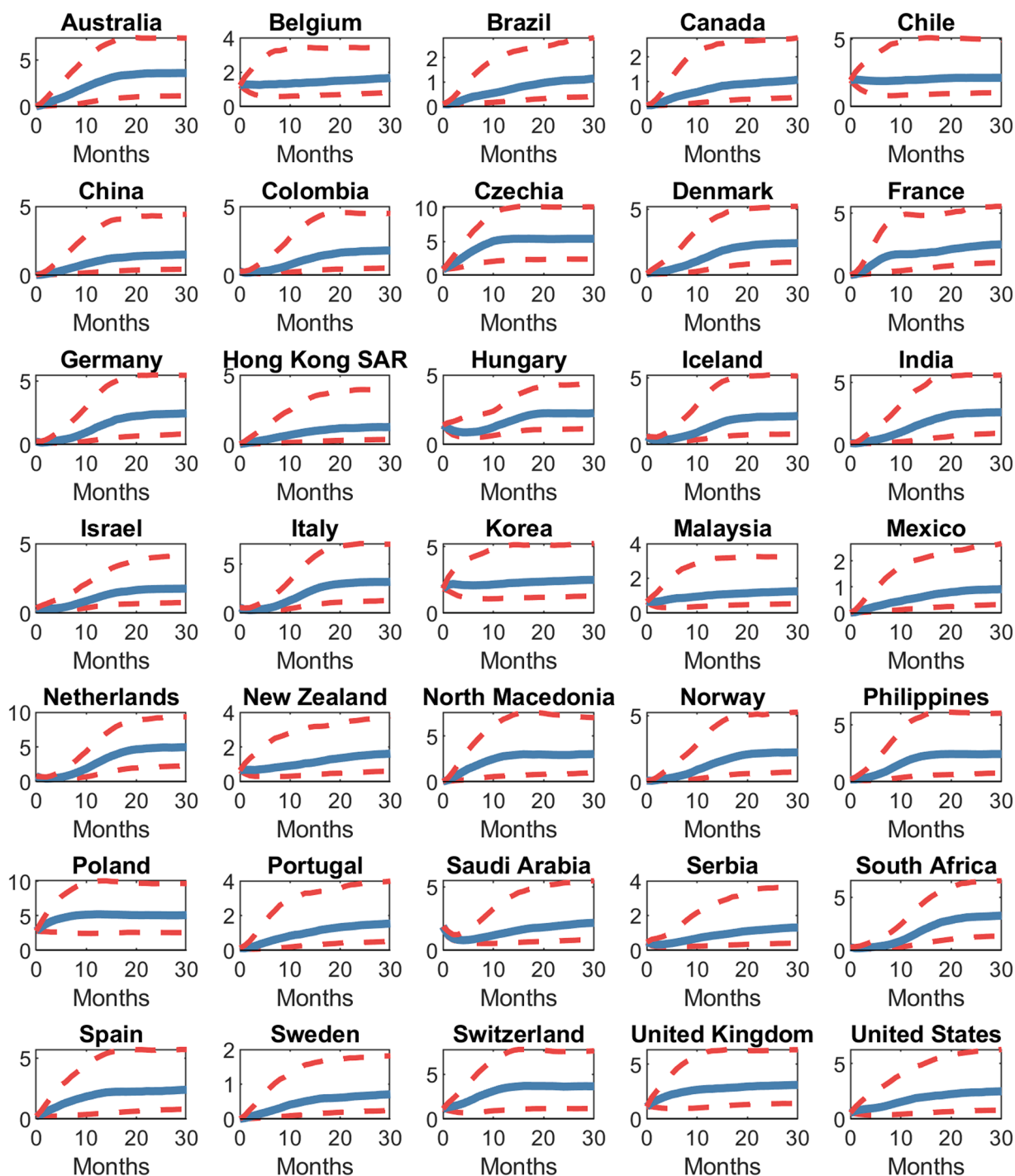


Fig. 2. Contribution of geopolitical risks to exchange rates. Notes: Figures represent the percentage contribution of geopolitical risks to exchange rate volatilities based on the forecast error variance decomposition. Solid lines represent the median across 1000 draws, whereas dotted lines represent the 68 % credible sets.

decreases with the horizon. This result is consistent with the view that geopolitical risks can trigger capital outflows and trade disruptions, both of which can depreciate exchange rates, and these effects are amplified for countries that are deeply integrated into global value chains due to their heightened reliance on foreign investment and international trade.

The relationship between the global value chain participation of countries and the contribution of geopolitical risks to exchange rate volatilities is given in Fig. 4 for alternative horizons after the shock. As is evident, the exchange rate volatilities of countries with

Table 2
Percentage contribution of geopolitical risks to exchange rate volatilities.

Country Information			Percentage Contribution of Geopolitical Risks			
Country Name	Country Code	GVC Participation	Contemporaneous	After One Month	After Three Months	After One Year
Australia	AU	49.92 %	0.02 %	0.12 %	0.53 %	2.55 %
Belgium	BE	72.28 %	1.23 %	1.29 %	1.26 %	1.35 %
Brazil	BR	33.98 %	0.10 %	0.08 %	0.19 %	0.63 %
Canada	CA	42.79 %	0.05 %	0.05 %	0.16 %	0.69 %
Chile	CL	37.53 %	1.94 %	2.00 %	1.93 %	1.97 %
China	CN	37.34 %	0.01 %	0.03 %	0.15 %	1.00 %
Colombia	CO	28.86 %	0.27 %	0.17 %	0.22 %	1.00 %
Czechia	CZ	57.20 %	0.82 %	1.25 %	2.14 %	5.29 %
Denmark	DK	59.83 %	0.12 %	0.19 %	0.28 %	1.36 %
France	FR	53.53 %	0.01 %	0.06 %	0.44 %	1.68 %
Germany	DE	56.73 %	0.22 %	0.14 %	0.22 %	1.33 %
Hong Kong SAR	HK	71.52 %	0.02 %	0.07 %	0.19 %	0.80 %
Hungary	HU	51.38 %	1.39 %	1.16 %	0.90 %	1.50 %
Iceland	IS	48.97 %	0.59 %	0.37 %	0.33 %	1.21 %
India	IN	50.19 %	0.11 %	0.08 %	0.19 %	1.31 %
Israel	IL	47.58 %	0.35 %	0.30 %	0.31 %	1.08 %
Italy	IT	47.40 %	0.55 %	0.34 %	0.32 %	1.75 %
Korea	KR	51.46 %	1.83 %	2.12 %	2.13 %	2.19 %
Malaysia	MY	58.58 %	0.65 %	0.64 %	0.70 %	1.01 %
Mexico	MX	39.28 %	0.01 %	0.03 %	0.15 %	0.54 %
Netherlands	NL	68.17 %	0.78 %	0.50 %	0.45 %	2.63 %
New Zealand	NZ	33.62 %	0.64 %	0.68 %	0.67 %	0.99 %
North Macedonia	MK	53.87 %	0.01 %	0.16 %	0.77 %	2.78 %
Norway	NO	57.73 %	0.08 %	0.06 %	0.17 %	1.17 %
Philippines	PH	53.03 %	0.19 %	0.19 %	0.33 %	1.96 %
Poland	PL	59.41 %	2.79 %	3.37 %	4.19 %	5.22 %
Portugal	PT	49.99 %	0.12 %	0.11 %	0.24 %	0.91 %
Saudi Arabia	SA	41.86 %	1.87 %	1.21 %	0.82 %	1.33 %
Serbia	RS	60.44 %	0.46 %	0.38 %	0.33 %	0.79 %
South Africa	ZA	42.88 %	0.28 %	0.18 %	0.22 %	1.30 %
Spain	ES	52.14 %	0.22 %	0.36 %	0.79 %	2.04 %
Sweden	SE	59.98 %	0.00 %	0.02 %	0.10 %	0.47 %
Switzerland	CH	47.06 %	1.11 %	1.28 %	1.54 %	3.42 %
United Kingdom	GB	54.43 %	1.10 %	1.49 %	1.91 %	2.75 %
United States	US	39.73 %	0.54 %	0.73 %	0.89 %	1.75 %

Notes: GVC Participation stands for the percentage participation in the global value chains. The estimates represent the percentage contribution of geopolitical risks to exchange rate volatilities based on the forecast error variance decomposition. The estimations are achieved for each country separately using the structural vector autoregression model for the sample period between January 2000 and June 2024.

higher global value chain participation are affected more by geopolitical shocks, especially in the long run, although this relationship is not as strong as the one represented in Fig. 3 based on the correlation coefficients. This result can be explained by global value chains magnifying the effects of geopolitical risks on exchange rate volatility, as they increase a country's exposure to sudden shifts in global capital flows and trade disruptions following geopolitical shocks.

3.3. Robustness checks

For robustness, two alternative measures of geopolitical events, namely geopolitical acts and geopolitical threats, are used in the estimations, where they are both borrowed from Caldara and Iacoviello (2022). The purpose of this robustness check is to understand the reasons behind the results above obtained by geopolitical risks that can be decomposed into geopolitical acts versus geopolitical threats based on the frequencies of alternative texts found in newspapers. Specifically, geopolitical acts are measured by the categories of beginning of war, escalation of war, and terror acts, whereas geopolitical threats are measured by the categories of war threats, peace threats, military buildups, nuclear threats, and terror threats. The corresponding results are given in the Online Appendix tables and figures, where it is suggested that geopolitical acts are effective on the exchange rates of more countries compared to geopolitical threats when statistically significant results are considered after one year. When the heterogeneity across countries is investigated, global value chain participation is more effective both on the cumulative impulse responses of exchange rates to geopolitical acts and the contribution of geopolitical acts to exchange rate volatilities. It is implied that while geopolitical threats can create some uncertainty in currency markets, the actual implementation of those threats, such as through military actions or the imposition of sanctions, has a more pronounced effect on exchange rates. This result can be explained by concrete actions triggering immediate and tangible consequences for international trade, investment flows, and investor confidence, leading to more significant changes in exchange rates.

For additional robustness, following Aliu and Bajra (2023), the effects of the Russian ruble (represented as percentage depreciation of the nominal effective exchange rate) on other exchange rates are investigated. The purpose of this robustness check is to understand

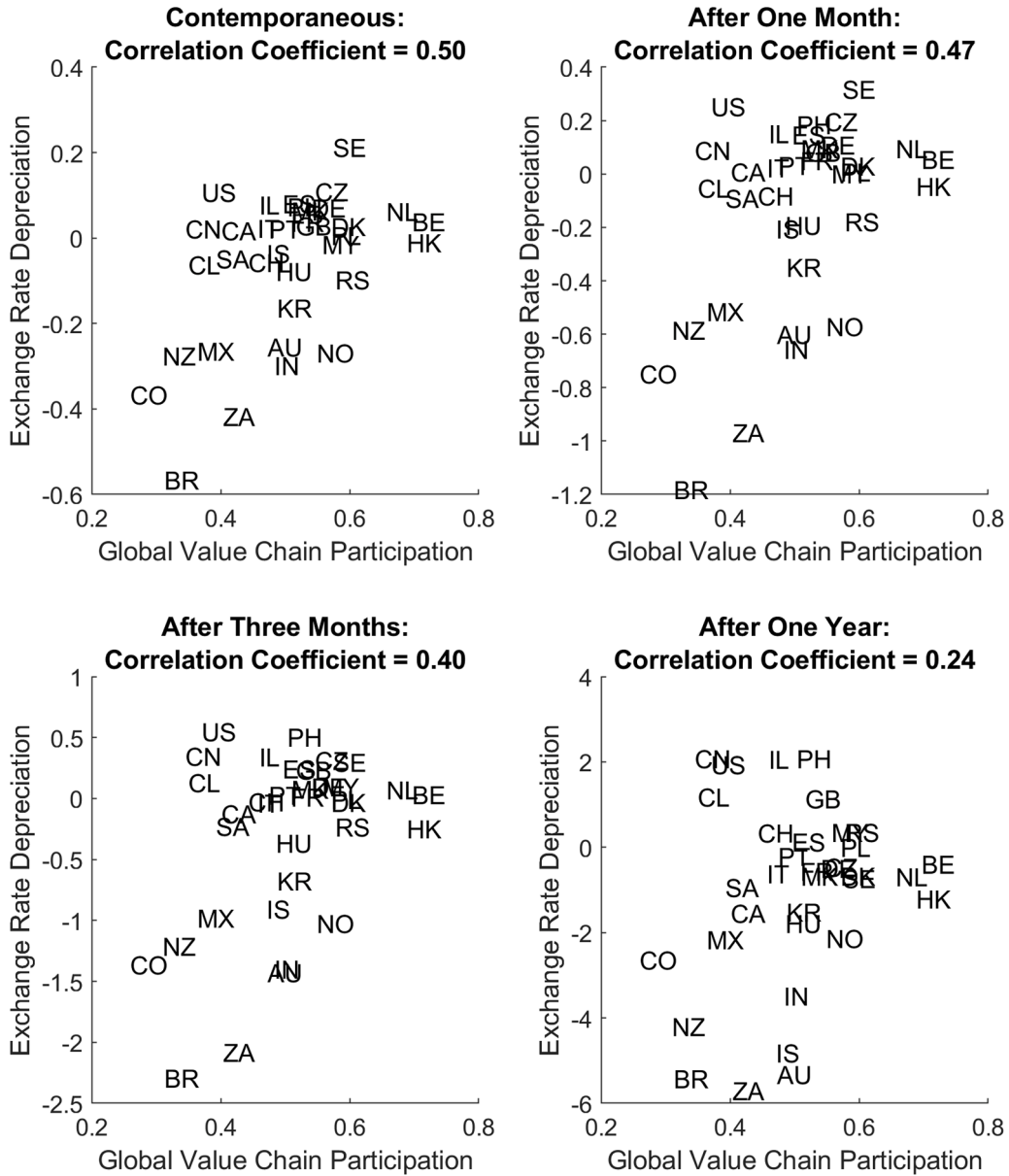


Fig. 3. Global value chains and effects of geopolitical risks. Notes: Figures represent the relationship between the percentage participation in global value chains and the cumulative impulse responses of exchange rates (represented as percentage depreciation) to one standard deviation shocks on global geopolitical risks.

the potential impact of the Russo-Ukraine War on the exchange rate of other countries. This is achieved by including the Russian ruble as an additional variable to the SVAR model, where it is ordered right before the policy rates. The corresponding results are given in the Online Appendix tables and figures, where it is suggested that the depreciation of the Russian ruble is associated with the depreciation of the nominal effective exchange rates in China, Hong Kong SAR, New Zealand, South Africa, Switzerland, and the United States. In comparison, the same depreciation of the Russian ruble is associated with the appreciation of the nominal effective exchange rates in Chile, Malaysia, Mexico, and the Philippines. When the heterogeneity across countries is investigated, the exchange rates of countries with higher global value chain participation depreciate less following the depreciation of the Russian ruble, especially in the short run, suggesting that the diversified trade relationships of these countries make them less susceptible to the ripple effects of a single currency's decline.

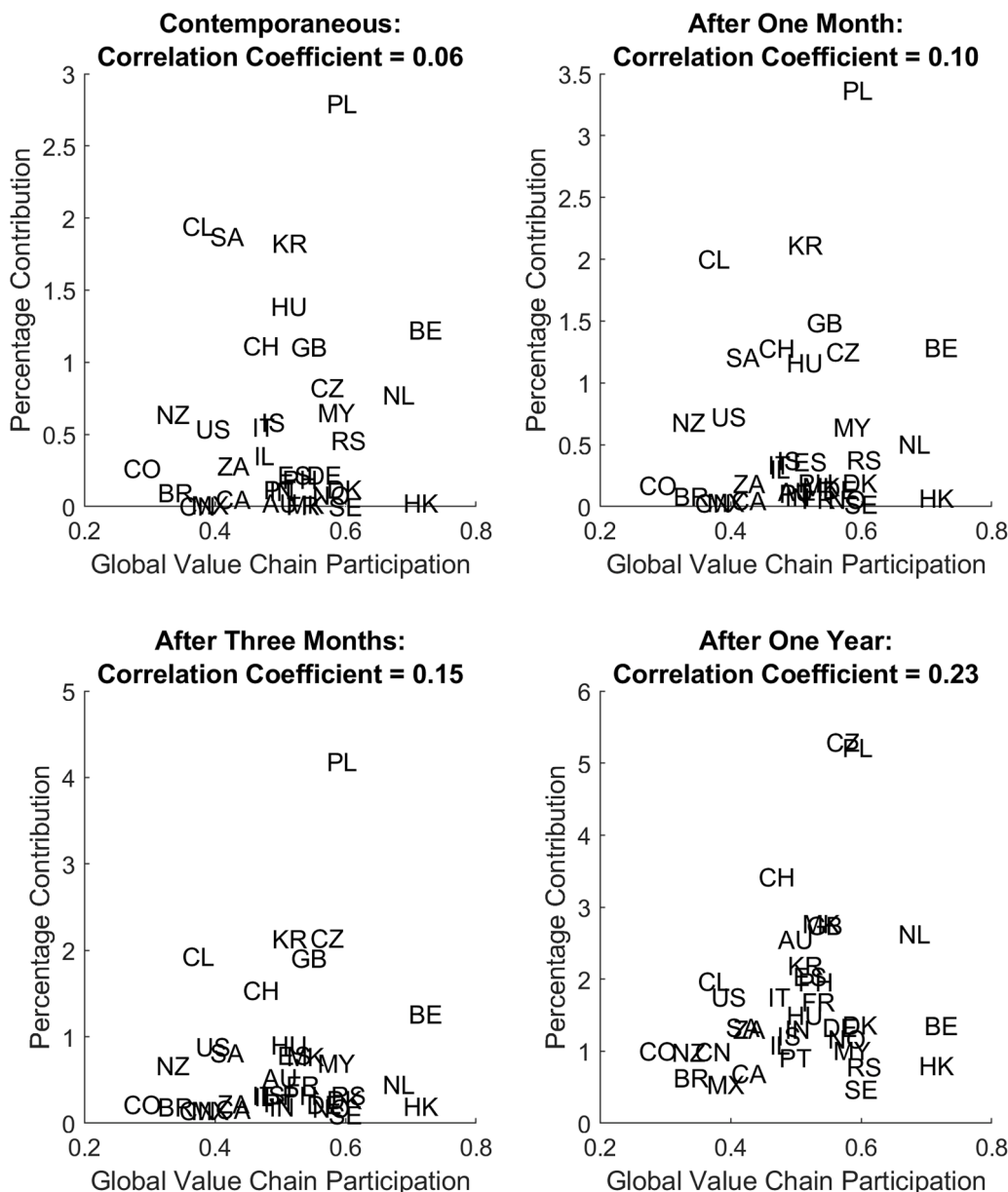


Fig. 4. Global value chains and contribution of geopolitical risks. Notes: Figures represent the relationship between the percentage participation in global value chains and the percentage contribution of geopolitical risks to exchange rate volatilities based on the forecast error variance decomposition.

4. Concluding remarks and policy suggestions

The results of this paper highlight the importance for policymakers, especially in countries deeply integrated into global value chains, to closely monitor and proactively address geopolitical risks to mitigate the corresponding effects on their exchange rates. Since geopolitical acts have more significant effects on exchange rates compared to geopolitical threats, policymakers may want to prioritize strategies that could reduce the likelihood of such events occurring. This could involve diplomatic efforts to de-escalate tensions, economic diversification to reduce reliance on vulnerable trade relationships, and the development of robust financial safety nets to cushion against geopolitical shocks. Furthermore, given the diverse responses of exchange rates to geopolitical risks across countries, policymakers may want to tailor their interventions to their specific circumstances, especially considering their level of global value chain participation.

Future research can delve deeper into the specific types of geopolitical risks and their varying impacts on exchange rates, particularly for countries embedded in global value chains. This may include investigating transmission mechanisms, the role of a

country's position within those value chains, and the effectiveness of different policy interventions like diplomacy, diversification, and financial safety nets. Expanding the scope to include long-term effects and interconnectedness with other economies would provide a more comprehensive understanding of this complex relationship.

CRedit authorship contribution statement

Hakan Yilmazkuday: Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.frl.2025.106769](https://doi.org/10.1016/j.frl.2025.106769).

Data availability

Data will be made available on request.

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